

# School of Natural Sciences and Mathematics

## Biology and Healthcare Management (Double Major) (BS)

### Bachelor of Science in Biology and Healthcare Management (Double Major)

[Degree Requirements](#) (148-150 semester credit hours)<sup>1</sup>

## JSOM Faculty

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**Professors:** Ashiq Ali, Gary Bolton, Huseyin Cavusoglu, Jianqing Chen, William M. Cready, Gregory G. Dess, Umit G. Gurun, Kyle Hyndman, Varghese S. Jacob, Dmitri Kuksov, Nanda Kumar, Seung-Hyun Lee, Zhiang (John) Lin, Sumit K. Majumdar, Syam Menon, Vijay S. Mookerjee, B. P. S. Murthi, Vikram Nanda, Mike W. Peng, Hasan Pirkul, Suresh Radhakrishnan, Srinivasan Raghunathan, Ram C. Rao, Brian Ratchford, Michael J. Rebello, Gil Sadka, Sumit Sarkar, Wing Kwong (Eric) Tsang, Jun Xia, Ying Xie, Harold Zhang, Zhiqiang (Eric) Zheng

**Associate Professors:** Mehmet Ayvaci, Nina Baranchuk, Zhonglan Dai, Rebecca Files, Surya N. Janakiraman, Robert L. Kieschnick Jr., Atanu Lahiri, Jun Li, Ningzhong Li, Livia Markóczy, Ramachandran (Ram) Natarajan, Naim Bugra Ozel, Young U. Ryu, Harpreet Singh, Upender Subramanian, Kelsey D. Wei, Han (Victor) Xia, Yexiao Xu, Alejandro Zentner, Jieying Zhang, Zhe (James) Zhang, Feng Zhao, Yibin Zhou

**Assistant Professors:** Sheen Levine, Radha Mookerjee, Alejandro Rivera Mesias, Christian Von Drathen, Guihua Wang, Steven Xiao

**Associate Professors Emeriti:** J. Richard Harrison, Jane Salk, David J. Springate

**Clinical Professors:** John Barden, Britt Berrett, Abhijit Biswas, Shawn Carraher, Larry Chasteen, Howard Dover, Randall S. Guttery, William Hefley, Marilyn Kaplan, Peter Lewin, Jeffrey Manzi, John McCracken, Daniel Rajaratnam, Kannan Ramanathan, Mark Thouin, Habte Woldu, Fang Wu

**Clinical Associate Professors:** Shawn Alborz, Dawn Owens, Carolyn Reichert

**Clinical Assistant Professors:** Moran Blueshtein, Jeffery (Jeff) Hicks, Liping Ma, Ravi Narayan, Parneet Pahwa, Nassim Sohaee

**Professors of Instruction:** Semiramis Amirpour, Mary Beth Goodrich, Chris Linsteadt, Matt Polze, Luell (Lou) Thompson

**Associate Professors of Instruction:** Judd Bradbury, Amal El-Ashmawi, Ayfer Gurun, Maria Hasenhuttl, Julie Haworth, Thomas (Tom) Henderson, Hubert Zydorek

**Assistant Professors of Instruction:** Daniel Karnuta, Joseph Mauriello, Victoria McCrady

**Professors of Practice:** Tiffany A. Bortz, Alexander Edsel, Rajiv Shah, Keith Thurgood

**Associate Professors of Practice:** Richard Bowen, Jackie Kimzey, Margaret Smallwood, Steven Solcher, Kathy Zolton

**Assistant Professors of Practice:** Edward Meda, Timothy Stephens, Robert Wright

## NSM Faculty

FACG> nsm-biology-and-healthcare-management-bs

**Professors:** Juan E. González, Tae Hoon Kim, Kelli Palmer, Lawrence J. Reitzer, Donal Skinner, Stephen Spiro, Li Zhang, Michael Qiwei Zhang

**Associate Professors:** Joseph Boll, Nicole De Nisco, Nikki Delk, Tian Hong, Faruck Morcos, Duane D. Winkler, Zhenyu Xuan

**Assistant Professors:** Nicholas Dillon, Xintong Dong, Lin Jia, Purna Joshi, Brandon Kim, Erica Sanchez, Darshan Sapkota, Yuki Shindo, Jon Sin

**Professors Emeriti:** Hans Bremer, Lee A. Bulla, Rockford Draper, Donald M. Gray

**Associate Professors Emeriti:** Gail A. M. Breen, Dennis L. Miller

**Clinical Professor:** David Murchison, Eberhard Voit

**Professors of Instruction:** Mehmet Candas, Elizabeth Pickett, Uma Srikanth

**Associate Professors of Instruction:** Meenakshi Maitra, Jing Pan, Eva Sadat, Subha Sarcar, Michelle Wilson, Zhuoru Wu, Wen-Ho Yu

**Assistant Professors of Instruction:** Stephanie Boyd, Andy Cheshire, Anne Davenport, Matthew Esparza, Amy Jo Gomez, Yi Huang, Li Liu, Simbarashe Mazambani, Iti Mehta, Ritu Mishra, Ramesh Padmanabhan, Narges Salamat

**Research Associate Professor:** Ru-Hung Wang

**Lecturer:** Kathleen McRoy

**Students must meet all course prerequisites.**

**I. Core Curriculum Requirements: 42 semester credit hours**

Curriculum Requirements can be fulfilled by other approved courses from institutions of higher education. The courses listed are recommended as the most efficient way to satisfy both Core Curriculum and Major Requirements at UT Dallas.

**Communication: 6 semester credit hours**

Select any 6 semester credit hours from [Communication Core](#) courses (see advisor)

**Mathematics: 3 semester credit hours**

[MATH 2413](#) Differential Calculus<sup>2, 3, 4, 5, 6</sup>

Or select any 3 semester credit hours from [Mathematics Core](#) courses<sup>7</sup> (see advisor)

**Life and Physical Sciences: 6 semester credit hours**

[CHEM 1311](#) General Chemistry I<sup>3</sup>

or [CHEM 1315](#) Honors Freshman Chemistry I<sup>3</sup>

[CHEM 1312](#) General Chemistry II<sup>3</sup>

or [CHEM 1316](#) Honors Freshman Chemistry II<sup>3</sup>

Or select any 6 semester credit hours from [Life and Physical Sciences Core](#) courses (see advisor)

**Language, Philosophy and Culture: 3 semester credit hours**

Select any 3 semester credit hours from [Language, Philosophy and Culture Core](#) courses (see advisor)

**Creative Arts: 3 semester credit hours**

Select any 3 semester credit hours from [Creative Arts Core](#) courses (see advisor)

**American History: 6 semester credit hours**

Select any 6 semester credit hours from [American History Core](#) courses (see advisor)

**Government/Political Science: 6 semester credit hours**

Select any 6 semester credit hours from [Government/Political Science Core](#) courses (see advisor)

### **Social and Behavioral Sciences: 3 semester credit hours**

Choose one of the following:<sup>7</sup> <sub>\_</sub>

[BA 1310](#) Making Choices in Free Market Systems<sup>2, 3</sup> <sub>\_</sub>

[BA 1320](#) Business in a Global World<sup>2, 3</sup> <sub>\_</sub>

[ECON 2301](#) Principles of Macroeconomics<sup>2, 3</sup> <sub>\_</sub>

[ECON 2302](#) Principles of Microeconomics<sup>2, 3</sup> <sub>\_</sub>

Or select any 3 semester credit hours from [Social and Behavioral Sciences Core](#) courses (see advisor)

### **Component Area Option: 6 semester credit hours**

Choose two of the following:<sup>7</sup> <sub>\_</sub>

[MATH 2414](#) Integral Calculus<sup>3, 4, 8</sup> <sub>\_ \_</sub>

[BA 1310](#) Making Choices in Free Market Systems<sup>2, 3</sup> <sub>\_</sub>

or [BA 1320](#) Business in a Global World<sup>2, 3</sup> <sub>\_</sub>

or [ECON 2301](#) Principles of Macroeconomics<sup>2, 3</sup> <sub>\_</sub>

Or select any 6 semester credit hours from [Component Area Option Core](#) courses (see advisor)

II. Major Requirements: 88-90 semester credit hours

### **Business Major Preparatory Courses: 12 semester credit hours beyond Core Curriculum**

[ACCT 2301](#) Introductory Financial Accounting<sup>2</sup> <sub>\_</sub>

[ACCT 2302](#) Introductory Management Accounting<sup>2</sup> <sub>\_</sub>

[BLAW 2301](#) Business and Public Law<sup>2</sup> <sub>\_</sub>

[OPRE 3360](#) Managerial Methods in Decision Making Under Uncertainty

or [STAT 2332](#) Introductory Statistics for Life Sciences

or [STAT 3360](#) Probability and Statistics for Management and Economics

Choose two of the following if not completed in Core Curriculum:<sup>7</sup>

[BA 1310](#) Making Choices in Free Market Systems<sup>2, 3</sup>

[BA 1320](#) Business in a Global World<sup>2, 3</sup>

[ECON 2301](#) Principles of Macroeconomics<sup>2, 3</sup>

[ECON 2302](#) Principles of Microeconomics<sup>2, 3</sup>

### **Business Core Courses: 28 semester credit hours**

[BA 1105](#) Professional Development (JSOM first-time-in-college freshmen are required to take [BA 1105](#) in their first semester)

or [BA 3105](#) Professional Development (Transfer students and students new to JSOM are required to take [BA 3105](#) in their first semester)

[BCOM 3300](#) Business Communication

[FIN 3320](#) Business Finance

[IMS 3310](#) International Business

[ITSS 3300](#) Information Technology for Business

[BUAN 3301](#) AI in Business

[MKT 3300](#) Principles of Marketing

[OBHR 3310](#) Organizational Behavior

or [OBHR 3330](#) Human Resource Management in Business

[OPRE 3310](#) Supply Chain and Operations Management

### ***Experiential Learning***

A capstone course is required for all undergraduate JSOM students.

[HMGH 4395](#) Capstone Senior Project - Healthcare Management

or [BPS 4395](#) Capstone Senior Project - Business<sup>9</sup>

A practicum experience of at least 160 working hours is required, with registration the courses below.

[HMG 4090](#) Healthcare Management Internship<sup>10</sup>

Students are required to complete 100 hours of community engagement experience with registration in the courses below.

[BA 4095](#) Social Sector Engagement and Community Outreach Practicum<sup>11</sup>

**Biology Major Preparatory Courses: 20-21 semester credit hours beyond Core Curriculum**

[CHEM 1111](#) General Chemistry Laboratory I

or [CHEM 1115](#) Honors Freshman Chemistry Laboratory I

[CHEM 1112](#) General Chemistry Laboratory II

or [CHEM 1116](#) Honors Freshman Chemistry Laboratory II

[CHEM 1311](#) General Chemistry I<sup>3</sup>

or [CHEM 1315](#) Honors Freshman Chemistry I<sup>3</sup>

[CHEM 1312](#) General Chemistry II<sup>3</sup>

or [CHEM 1316](#) Honors Freshman Chemistry II<sup>3</sup>

[CHEM 2323](#) Introductory Organic Chemistry I<sup>2</sup>

or [CHEM 2327](#) Honors Organic Chemistry I<sup>2</sup>

[CHEM 2325](#) Introductory Organic Chemistry II<sup>2</sup>

or [CHEM 2328](#) Honors Organic Chemistry II<sup>2</sup>

[CHEM 2233](#) Introductory Organic Chemistry Laboratory<sup>2</sup>

or [CHEM 2237](#) Honors Organic Chemistry Laboratory<sup>2</sup>

[MATH 2413](#) Differential Calculus<sup>2, 3, 4, 5, 6</sup>

[MATH 2414](#) Integral Calculus<sup>3, 4, 8</sup>

[PHYS 2325](#) Mechanics

and [PHYS 2125](#) Physics Laboratory I

or [PHYS 2421](#) Honors Physics I - Mechanics and Heat<sup>12</sup>

or [PHYS 1301](#) College Physics I

and [PHYS 1101](#) College Physics Laboratory I

[PHYS 2326](#) Electromagnetism and Waves

and [PHYS 2126](#) Physics Laboratory II

or [PHYS 2422](#) Honors Physics II - Electromagnetism and Waves

and [PHYS 2126](#) Physics Laboratory II

or [PHYS 1302](#) College Physics II

and [PHYS 1102](#) College Physics Laboratory II

### **Biology Core Courses: 28-29 semester credit hours**

[BIOL 2111](#) Introduction to Modern Biology Workshop I<sup>2</sup>

[BIOL 2112](#) Introduction to Modern Biology Workshop II<sup>2</sup>

[BIOL 2281](#) Introductory Biology Laboratory<sup>2</sup>

[BIOL 2311](#) Introduction to Modern Biology I<sup>2</sup>

[BIOL 2312](#) Introduction to Modern Biology II<sup>2</sup>

[BIOL 3401](#) Genetics

[BIOL 3402](#) Molecular and Cell Biology

[BIOL 3461](#) Biochemistry I

or [CHEM 3461](#) Biochemistry I

[BIOL 3462](#) Biochemistry II

or [CHEM 3462](#) Biochemistry II

or [BIOL 3335](#) Microbial Physiology

[BIOL 3380](#) Biochemistry Laboratory

### **III. Elective Requirements: 18 semester credit hours**

## **Guided Electives: 18 semester credit hours**

## **Healthcare Management Core Courses: 15 semester credit hours**

[HMG1 3301](#) Healthcare Management

[HMG1 3310](#) Healthcare Regulatory Environment

[HMG1 3311](#) Healthcare Financial Analysis

[HMG1 3373](#) Healthcare Database Management

[HMG1 4321](#) Healthcare Information Systems

## **Biology (3 semester credit hours):**

[BIOL 4380](#) Cell and Molecular Biology Laboratory<sup>13</sup>

or [BIOL 3V96](#) Undergraduate Research in Molecular and Cell Biology

or [BIOL 4391](#) Senior Research in Molecular and Cell Biology

or [BIOL 4399](#) Senior Honors Research for Thesis in Molecular and Cell Biology

The plan must include sufficient upper-division courses to total 45 upper-division semester credit hours.

1. Degree is 148-150 semester credit hours if students are required to take NATS 1101.
2. Indicates a prerequisite class to be completed before enrolling for upper-division classes.
3. A required Major course that also fulfills a Core Curriculum requirement. Semester credit hours are counted in Core Curriculum.
4. Six semester credit hours of Calculus are counted under Mathematics Core and Component Area Option Core, and 2 semester credit hours of Calculus are counted as Biology Major Preparatory Courses.
5. MATH 2413 is required for Healthcare Management/Biology prescribed double majors. Students may elect to substitute MATH 2417 for MATH 2413.
6. To make timely degree progress, students should complete MATH 2413 or MATH 2417 by the end of their first semester at UT Dallas. Students who will not meet this requirement should contact their academic advisor to discuss their degree timeline.
7. Certain courses listed are prerequisites for major core, major concentration, or major related courses. Choose accordingly.



8. Students may elect to substitute MATH 2419 for MATH 2414.
9. BPS 4395 can be taken instead of HMGT 4395 only if HMGT 4395 is not available.
10. Students may fulfill the internship requirement with HMGT 4090, BA 4090, or HMGT 4V90 (1-3 semester credit hours). The zero semester credit hour courses HMGT 4090 or BA 4090 are is recommended as the most efficient way to satisfy this requirement.
11. Students may fulfill the community engagement requirement with BA 4095, BPS 4396, ENTP 4340, or MKT 4360, or OPRE 4391. The zero semester credit hour course BA 4095 is recommended as the most efficient way to satisfy this requirement.
12. Students who complete PHYS 2421 do not need to complete PHYS 2125.
13. Requires permission of the Biology Undergraduate Advisor to ensure training in recombinant DNA analysis.

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